

HOUSTON SOUTHWEST, TX, USA

WMO: 720637

Lat: 29.500N

Lon: 95.477W

Elev: 21

StdP: 101.07

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00223

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	0.6	-9.9	1.6	4.7	-6.9	2.1	6.2	9.4	11.3	8.4	11.7	1.7	350	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.0	36.0	24.1	34.9	24.6	33.8	24.8	27.1	30.6	26.7	30.7	26.3	30.5	2.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	27.9	25.7	21.0	27.8	25.2	20.4	27.7	85.0	30.7	83.0	31.1	81.5	30.5	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.7	6.8	5.7	-4.4	37.3	2.3	1.5	-6.1	38.4	-7.4	39.2	-8.8	40.1	-10.5	41.1	
			DB												
			WB	-5.5	27.5	2.5	0.5	-7.2	27.9	-8.7	28.2	-10.1	28.4	-11.9	28.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	11.2	14.3	17.6	20.7	24.2	27.3	27.9	28.5	26.2	21.4	16.2	12.8
	DBStd	7.07	5.19	5.52	4.38	3.50	2.88	1.66	1.57	1.36	2.07	3.99	5.22	5.19
	HDD10.0	109	47	21	4	0	0	0	0	0	0	0	9	28
	HDD18.3	752	225	132	68	18	3	0	0	0	0	17	102	187
	CDD10.0	4018	86	142	238	320	439	518	554	573	484	354	194	114
	CDD18.3	1619	6	20	44	88	184	268	296	315	234	112	37	14
	CDH23.3	15183	21	77	210	565	1592	2775	3145	3420	2083	1010	223	63
	CDH26.7	6503	0	5	23	94	539	1314	1513	1743	882	345	41	3
Wind	WSAvg	2.2	2.6	2.8	2.7	2.8	2.5	1.8	1.4	1.5	1.6	1.9	2.2	2.5
Precipitation	PrecAvg	1352	108	76	94	103	132	148	115	109	119	136	107	105
	PrecMax	2020	267	231	215	250	335	625	308	231	291	446	322	210
	PrecMin	602	15	13	20	2	1	7	35	13	41	1	18	28
	PrecStd	331	60	56	54	77	84	121	73	66	59	114	70	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.4	27.6	28.6	30.7	33.7	36.6	36.4	37.4	35.3	32.7	30.0	27.1
		MCWB	18.9	20.5	20.3	22.4	23.3	23.4	24.0	23.9	23.5	23.6	22.1	20.2
	2%	DB	23.7	25.7	27.2	28.9	32.2	35.0	35.2	36.2	33.8	31.5	27.8	25.2
		MCWB	18.1	19.8	20.4	21.7	23.5	24.2	24.6	24.5	24.2	24.0	21.6	20.7
	5%	DB	21.9	23.9	25.8	27.7	31.0	33.6	34.1	35.0	32.6	30.2	26.0	23.2
		MCWB	17.7	19.4	20.0	21.3	23.4	24.6	24.9	25.0	24.2	23.5	21.0	20.3
	10%	DB	20.1	22.2	24.1	26.6	29.9	32.6	33.0	33.9	31.4	28.8	24.0	21.9
		MCWB	17.0	19.1	19.1	20.7	23.1	24.8	25.1	25.2	24.3	22.4	20.1	19.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	22.1	22.7	25.1	25.8	27.2	27.3	27.6	27.2	28.8	24.5	23.0
		MCDB	22.4	24.8	26.0	28.1	29.8	31.1	30.5	31.2	29.0	29.5	27.6	24.1
	2%	WB	20.3	21.4	21.9	23.9	25.3	26.6	26.8	27.1	26.4	25.7	23.4	22.1
		MCDB	21.9	23.7	25.0	26.7	28.9	30.7	30.7	31.3	29.7	29.2	25.6	23.7
	5%	WB	19.4	20.8	21.4	23.2	24.8	26.1	26.4	26.6	25.9	24.8	22.6	21.2
		MCDB	21.3	22.5	23.9	25.7	28.4	30.3	30.9	31.1	29.5	28.4	24.5	23.0
	10%	WB	17.6	20.1	20.8	22.5	24.2	25.7	26.0	26.2	25.5	23.8	21.6	19.9
		MCDB	19.0	21.7	22.8	24.9	27.9	29.9	30.4	30.8	29.2	26.9	23.6	21.6
Mean Daily Temperature Range	5% DB	MDBR	11.4	10.3	10.7	10.8	10.6	10.5	10.5	11.0	10.1	12.2	11.8	10.5
		MCDBR	13.0	10.9	11.0	11.0	11.3	12.2	12.2	12.8	12.0	12.4	12.1	12.0
		MCWBR	8.5	6.1	5.6	5.4	4.0	3.7	3.8	3.8	4.2	5.7	6.7	8.1
	5% WB	MCDBR	10.0	9.0	8.9	8.4	9.0	10.1	10.6	11.1	9.5	10.6	9.8	10.9
		MCWBR	8.1	6.2	5.3	5.0	3.7	3.9	3.9	4.0	3.9	5.7	6.3	8.3
Clear-Sky Solar Irradiance	taub	0.344	0.360	0.376	0.438	0.482	0.469	0.496	0.484	0.455	0.381	0.374	0.352	
	taud	2.491	2.456	2.402	2.229	2.141	2.233	2.174	2.210	2.258	2.457	2.424	2.475	
	Ebn at Noon	891	906	911	861	820	827	802	810	822	872	848	862	
	Edh at Noon	92	103	116	142	156	141	150	143	132	102	97	89	
All-Sky Solar Radiation	RadAvg	2.88	3.25	4.44	5.28	5.88	6.08	5.92	5.70	4.95	4.34	3.27	2.56	
	RadStd	0.28	0.56	0.37	0.39	0.43	0.50	0.59	0.48	0.53	0.51	0.33	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)		+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+99

Nomenclature: See separate page